

Topic 7: The z-Transform

[Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder. *Signal Processing First*. Prentice Hall, 2003.]

Topics: definition and properties of the z-transform, system functions, time shifting, convolution and polynomial multiplication, cascaded systems, factorisation and inverse filtering, the unit circle, poles and zeros, nulling filters, running-sum and band-pass filters, and linear-phase FIR systems.

Key idea. The z-transform converts time-domain operations into algebra: delays become powers of z^{-1} , convolution becomes multiplication, and the poles and zeros of a system function reveal how the filter behaves in frequency.

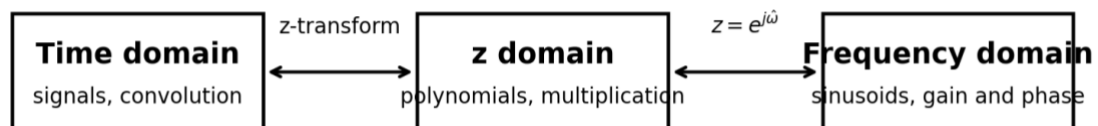


Figure 7.0. Relationship among the time, z, and frequency domains.

7.1 Definition of the z-Transform

A finite-length signal $x[n]$ can be represented by the relation

$$x[n] = \sum_{k=0}^N x[k] \delta[n - k]$$

and the z-transform of this signal is defined by

$$X(z) = \sum_{k=0}^N x[k] z^{-k}$$

where z may be any complex number; that is, z is the independent variable of the z-transform. The transform may also be written as

$$X(z) = \sum_{k=0}^N x[k] (z^{-1})^k$$

which emphasizes that $X(z)$ is simply a polynomial of degree N in the variable z^{-1} .

To calculate the z-transform, construct a polynomial whose coefficients are the values of the sequence $x[n]$. The inverse process is equally direct. We emphasize this correspondence using the relation

$$\begin{array}{ccc} n \text{ domain} & & z \text{ domain} \\ x[n] = \sum_{k=0}^N x[k] \delta[n - k] & \Leftrightarrow & X(z) = \sum_{k=0}^N x[k] z^{-k} \end{array}$$

A simple example uses a shifted impulse in the n domain.

$$\begin{array}{ccc} n \text{ domain} & & z \text{ domain} \\ x[n] = \delta[n - n_0] & \Leftrightarrow & X(z) = z^{-n_0} \end{array}$$

7.2 The z-Transform and Linear Systems

The z-transform is indispensable in the design and analysis of LTI systems.

7.2.1 The z-Transform of an FIR Filter

Recall the representations of an FIR filter.

$$y[n] = \sum_{k=0}^M b_k x[n-k] = x[n] * h[n]$$

The impulse response $h[n]$ can be represented as

$$h[n] = \sum_{k=0}^M b_k \delta[n-k]$$

Consider the signal

$$x[n] = z^n \quad \text{for all } n,$$

where z is any complex number. The corresponding output is

$$\begin{aligned} y[n] &= \sum_{k=0}^M b_k z^{n-k} \\ &= \left(\sum_{k=0}^M b_k z^{-k} \right) z^n \end{aligned}$$

The term in parentheses is called the system function, or transfer function, of the FIR filter. It is the z-transform of the impulse response. Therefore, the system function of an FIR filter is defined as

$$H(z) = \sum_{k=0}^M b_k z^{-k} = \sum_{k=0}^M h[k] z^{-k}$$

7.3 Properties of the z-Transform

7.3.1 Superposition Property

The z-transform is a linear transformation.

$$ax_1[n] + bx_2[n] \Leftrightarrow aX_1(z) + bX_2(z)$$

7.3.2 Time-Shifting Property

Multiplication by z^{-1} in the z domain corresponds to a delay of one sample in the n domain.

$$x[n-1] \Leftrightarrow z^{-1}X(z)$$

This generalizes to

Example. For $x[n]=\{3,1,4,1,5,9,0,\dots\}$, the z-transform is $X(z)=3+z^{-1}+4z^{-2}+z^{-3}+5z^{-4}+9z^{-5}$. Delaying the sequence by one sample multiplies its transform by z^{-1} .

$$\begin{aligned} X(z) &= 3 + z^{-1} + 4z^{-2} + z^{-3} + 5z^{-4} + 9z^{-5} \\ x[n-1] &\Leftrightarrow z^{-1}X(z) = 3z^{-1} + z^{-2} + 4z^{-3} + z^{-4} + 5z^{-5} + 9z^{-6} \\ x[n-n_0] &\Leftrightarrow z^{-n_0}X(z) \end{aligned}$$

7.3.3 A General Formula for the z-Transform

The finite-length definition

$$X(z) = \sum_{n=0}^N x[n]z^{-n}$$

extends to infinite-length signals as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

7.4 The z-Transform as an Operator

Because z^{-1} represents a one-sample delay in the n domain, it is often interpreted as an operator. This interpretation is useful but can be confusing.

7.4.1 Unit-Delay Operator

The unit-delay system is one of the basic building blocks of an FIR difference equation.

$$y[n] = D\{x[n]\} = x[n - 1]$$

The z-transform description of the system is obtained by using the input $x[n]=z^n$ for all n , where z is a complex number. The output is then

$$y[n] = D\{z^n\} = z^{n-1} = z^{-1}z^n = z^{-1}x[n]$$

This expression is strictly valid only for the input $x[n]=z^n$, but it is common to use z^{-1} in place of the delay operator D .

7.4.2 Operator Notation

Consider the system

$$y[n] = x[n] - x[n - 1]$$

The operator describing the system is $(1 - z^{-1})$, because the system can be written as

$$y[n] = (1 - z^{-1})x[n]$$

7.4.3 Operator Notation in Block Diagrams

The unit-delay operator is particularly useful in block-diagram representations of LTI systems.

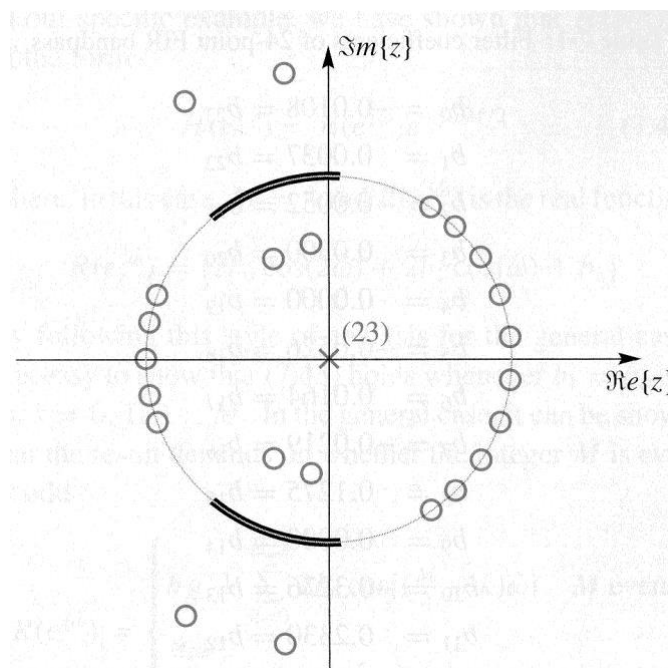


Figure 7.1. Computational structure of a first-order FIR filter. (a) Equivalence between z^{-1} and a unit delay. (b) Block diagram of a first-order filter.

7.5 Convolution and the z-Transform

The impulse response of a unit-delay system is

$$h[n] = \delta[n - 1]$$

and therefore a one-sample delay is equivalent to convolution with a shifted impulse.

$$y[n] = x[n] * \delta[n - 1] = x[n - 1]$$

The system function of the unit-delay system is the z-transform of its impulse response,

$$H(z) = z^{-1}$$

and in the z domain the output can be written as

$$Y(z) = z^{-1}X(z) = H(z)X(z)$$

To prove this equality, recall that

$$y[n] = x[n] * h[n] = \sum_{k=0}^M h[k]x[n - k]$$

Using superposition and time shifting, the z-transform of $y[n]$ is

$$\begin{aligned} Y(z) &= \sum_{k=0}^M h[k]z^{-k}X(z) \\ &= H(z)X(z) \end{aligned}$$

Thus, convolution in the n domain corresponds to multiplication in the z domain:

$$y[n] = h[n] * x[n] \quad \Leftrightarrow \quad Y(z) = H(z)X(z)$$

This also shows that convolution and polynomial multiplication are essentially the same operation.

Worked example. Let $x[n]=\{1,2,1\}$ and $h[n]=\{1,-1\}$. Their z-transforms are multiplied, and the coefficients of the product give the convolution output.

$$X(z) = 1 + 2z^{-1} + z^{-2}, \quad H(z) = 1 - z^{-1}$$

$$Y(z) = X(z)H(z) = 1 + z^{-1} - z^{-2} - z^{-3}$$

$$y[n] = \{1, 1, -1, -1\}$$

7.5.1 Cascaded Systems

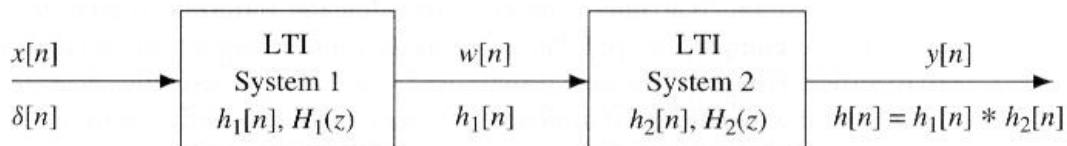


Figure 7.2. Cascade of two LTI systems.

The system function of two cascaded LTI systems is the product of the individual system functions.

$$h[n] = h_1[n] * h_2[n] \quad \Leftrightarrow \quad H(z) = H_1(z)H_2(z)$$

As an example, consider a system described by the two difference equations

$$w[n] = 3x[n] - x[n - 1]$$

$$y[n] = 2w[n] - w[n - 1]$$

The two filters can be combined into a single difference equation.

$$\begin{aligned} y[n] &= 2(3x[n] - x[n - 1]) - (3x[n - 1] - x[n - 2]) \\ &= 6x[n] - 5x[n - 1] + x[n - 2] \end{aligned}$$

In the z domain, the two filters are

$$H_1(z) = 3 - z^{-1}, \quad H_2(z) = 2 - z^{-1}$$

and the overall system function is

$$H(z) = (3 - z^{-1})(2 - z^{-1}) = 6 - 5z^{-1} + z^{-2}$$

7.5.2 Factorisation of z-Polynomials

Just as z-transforms can be multiplied to obtain higher-order systems, a z-transform polynomial can be factored to decompose a large system into smaller modules.

Consider the following example.

$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

One root of $H(z)$ is $z=1$, so $(1-z^{-1})$ is a factor of $H(z)$. The other factor is obtained by division.

$$H_2(z) = \frac{H(z)}{H_1(z)} = \frac{H(z)}{1 - z^{-1}} = 1 - z^{-1} + z^{-2}$$

Therefore, $H(z)$ can be written as

$$H(z) = (1 - z^{-1})(1 - z^{-1} + z^{-2})$$

7.5.3 Deconvolution

Can the second filter in a cascade undo the effect of the first? We want the output of the second filter to equal the input of the first. Suppose $H_1(z)$ is known and we seek $H_2(z)$ such that the overall system has unity response. The z-transform analysis requires

$$Y(z) = H_1(z)H_2(z)X(z) = H(z)X(z) = X(z)$$

Because the first system processes the input by convolution, the second filter undoes that convolution. This process is called deconvolution or inverse filtering. If $H_1(z)H_2(z)=1$, then $H_2(z)$ is the inverse of $H_1(z)$.

For example, if $H_1(z)$ is known and the desired overall response is $H(z)=1$, then

$$H(z) = H_1(z)H_2(z) = 1$$

we obtain

$$H_2(z) = \frac{1}{H_1(z)}$$

7.6 Relation between the z Domain and the Frequency Domain

Frequency domain		z domain
$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k}$	$\Leftrightarrow$	$H(z) = \sum_{k=0}^M b_k z^{-k}$

The correspondence between the two domains follows by substituting $z=e^{j\hat{\omega}}$ into $H(z)$.

$$H(e^{j\hat{\omega}}) = H(z)|_{z=e^{j\hat{\omega}}}$$

7.6.1 The z-Plane and the Unit Circle

From this point onward, we use $H(e^{j\hat{\omega}})$ to emphasize the connection between the normalized-frequency domain and the z domain. Values $z=e^{j\hat{\omega}}$ lie on the unit circle.

$$z = e^{j\hat{\omega}}, \quad |z| = 1$$

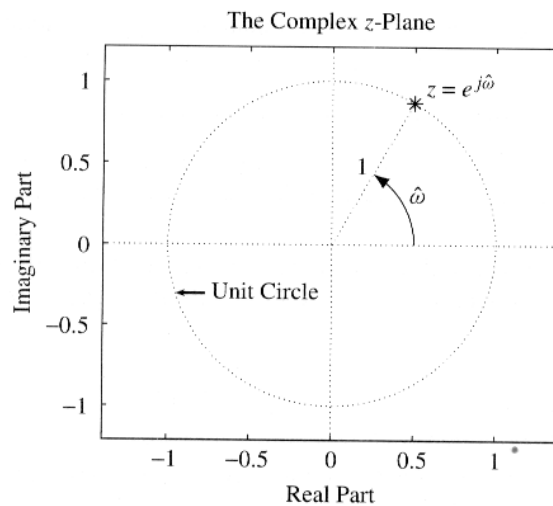


Figure 7.4. The complex z-plane with $z=e^{j\hat{\omega}}$.

Many properties of a system frequency response become clear in the z-plane. For example, its 2π periodicity follows directly from the unit-circle representation.

7.6.2 Zeros and Poles of $H(z)$

The system function of an FIR filter is described essentially by its zeros.

Consider a system function that can be expressed in several equivalent forms.

$$\begin{aligned} H(z) &= 1 - 2z^{-1} + 2z^{-2} - z^{-3} \\ &= (1 - z^{-1})(1 - e^{j\pi/3}z^{-1})(1 - e^{-j\pi/3}z^{-1}) \end{aligned}$$

Multiplying numerator and denominator by z^3 gives two additional equivalent forms.

$$H(z) = \frac{z^3 - 2z^2 + 2z - 1}{z^3} = \frac{(z - 1)(z - e^{j\pi/3})(z - e^{-j\pi/3})}{z^3}$$

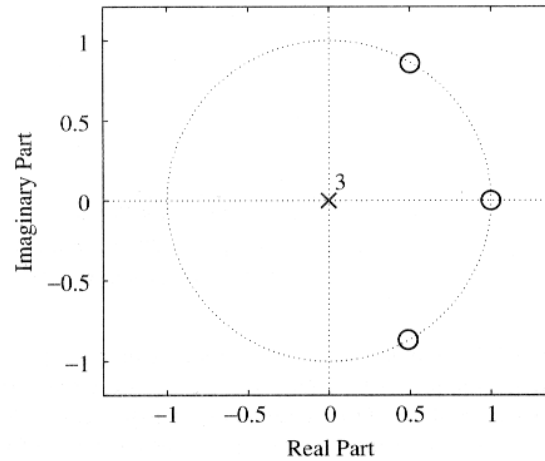
The final expression shows that the zeros of $H(z)$ are located at $z=1$ and $z=e^{\pm j\pi/3}$. It also shows that $H(z)$ becomes unbounded as z approaches zero. Values of z for which $H(z)$ is undefined are called poles; here $H(z)$ has a pole of order three at $z=0$. **$H(z)$**

The pole-zero locations are also evident in the factored form, because each factor of the form $(1-az^{-1})$ can be written as

$$1 - az^{-1} = \frac{z - a}{z}$$

Thus, each factor $(1-az^{-1})$ represents a zero at $z=a$ and a pole at $z=0$.

It is useful to display the poles and zeros of $H(z)$ as points in the complex z-plane.

Figure 7.5. Poles and zeros of $H(z)$ in the z -plane.

7.6.3 Importance of the Zeros of $H(z)$

The zeros of a polynomial system function determine $H(z)$, apart from an overall multiplicative constant.

For certain inputs, knowing the zero locations is sufficient to determine the output without explicitly using the difference equation. For an input $x[n] = z_0^n$, the output is

$$y[n] = H(z_0)z_0^n$$

The factor $H(z_0)$ is a complex constant that changes the input magnitude and phase. If z_0 is a zero of $H(z)$, then $H(z_0) = 0$ and the output is identically zero.

For the example system, the roots are

$$z_1 = 1, \quad z_2 = e^{j\pi/3}, \quad z_3 = e^{-j\pi/3}$$

These zeros lie on the unit circle and correspond to frequencies at which the system gain is zero. Complex sinusoids at those frequencies are therefore blocked or nulled.

7.6.4 Nulling Filters

The preceding result allows FIR filters to be designed to suppress specific sinusoidal components of an input signal.

Example. For the second-order nulling filter designed for $\omega_0 = \pi/4$, substituting $z = e^{j\pi/4}$ gives $H(e^{j\pi/4}) = 0$. Therefore an input $\cos(\pi n/4)$ is completely removed in steady state.

$$H(e^{j\pi/4}) = 1 - \sqrt{2}e^{-j\pi/4} + e^{-j\pi/2} = 0$$

$$x[n] = \cos(\pi n/4) \Rightarrow y[n] = 0$$

A zero in the z -plane suppresses a complex exponential of the form z_0^n . To suppress a real sinusoid, both complex exponentials in its Euler representation must be suppressed.

$$x[n] = \cos(\hat{\omega}_0 n) = 1/2 e^{j\hat{\omega}_0 n} + 1/2 e^{-j\hat{\omega}_0 n}$$

Each complex exponential can be suppressed with a first-order FIR filter. Cascading two such filters gives a second-order filter that suppresses the cosine. The two zeros are located at $e^{\pm j\hat{\omega}_0}$, and the system function is

$$\begin{aligned} H(z) &= (1 - e^{j\hat{\omega}_0} z^{-1})(1 - e^{-j\hat{\omega}_0} z^{-1}) \\ &= 1 - 2\cos(\hat{\omega}_0)z^{-1} + z^{-2} \end{aligned}$$

Figure 7.6 shows the pair of zeros required to remove components at $\omega_0 = \pm\pi/4$. The corresponding system function is

$$H(z) = 1 - 2\cos\left(\frac{\pi}{4}\right)z^{-1} + z^{-2} = 1 - \sqrt{2}z^{-1} + z^{-2}$$

and the difference equation is

$$y[n] = x[n] - \sqrt{2}x[n-1] + x[n-2]$$

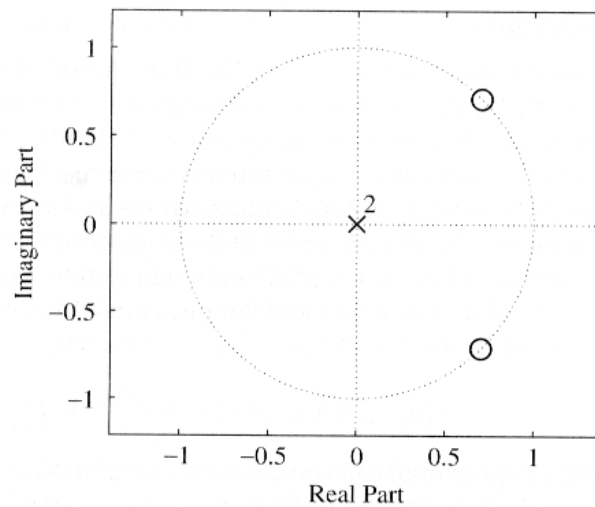


Figure 7.6. Unit-circle zeros of a second-order nulling filter for a sinusoid at $\omega_0 = \pm\pi/4$. The two poles are at the origin.

Redrawn view of the conjugate zeros that null a sinusoid at $\pm\pi/4$.

7.6.5 Graphical Relation between z and Frequency

The substitution $z = e^{j\omega}$ links the z domain and the normalized-frequency domain. The frequency response is obtained by evaluating the system function on the unit circle. Figure 7.7 shows $|H(z)|$ over a region of the z -plane that includes the unit circle.

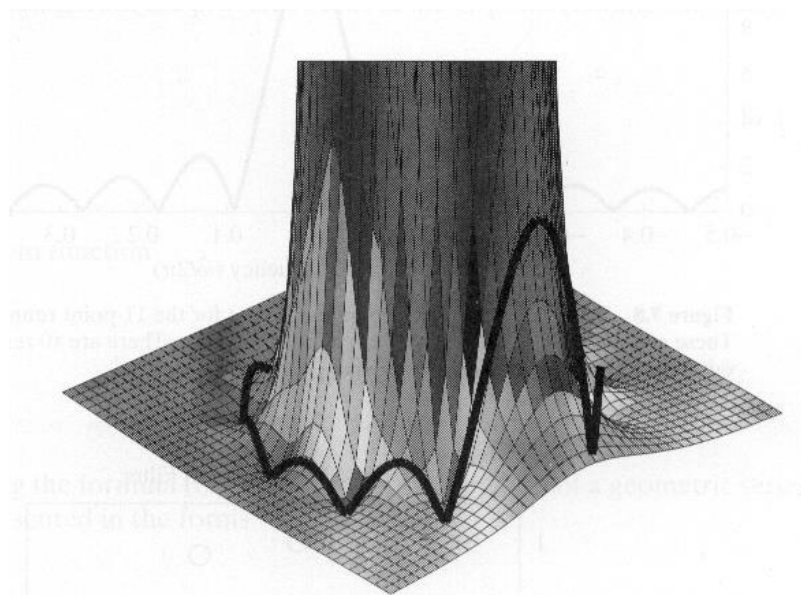


Figure 7.7. Magnitude of the z -transform of an FIR filter over a region of the z -plane. The black curve is the unit-circle trace and therefore the magnitude frequency response.

The system in this example is an 11-point running-sum filter whose coefficients are all equal to one. Its system function is

$$H(z) = \sum_{k=0}^{10} z^{-k}$$

7.7 Useful Filters

This section introduces band-pass filters related to running-sum filters.

7.7.1 A Running-Sum Filter

An L-point running-sum filter is defined by

$$y[n] = \sum_{k=0}^{L-1} x[n-k]$$

and has the system function

$$H(z) = \sum_{k=0}^{L-1} z^{-k} = \frac{1 - z^{-L}}{1 - z^{-1}}$$

The zeros of $H(z)$ are the roots of the numerator, namely the values of z that satisfy

$$z^L - 1 = 0 \quad \Leftrightarrow \quad z^L = 1$$

Because $e^{j2\pi k} = 1$, each value

$$z_k = e^{j2\pi k/L}, \quad k = 0, 1, \dots, L-1$$

satisfies $z^L = 1$ and is therefore a root. The apparent pole at $z=1$ is cancelled by the numerator zero, leaving only the poles at the origin introduced by $z^{-(L-1)}$. Hence $H(z)$ can be written as

$$H(z) = \prod_{k=1}^{L-1} (1 - e^{j2\pi k/L} z^{-1})$$

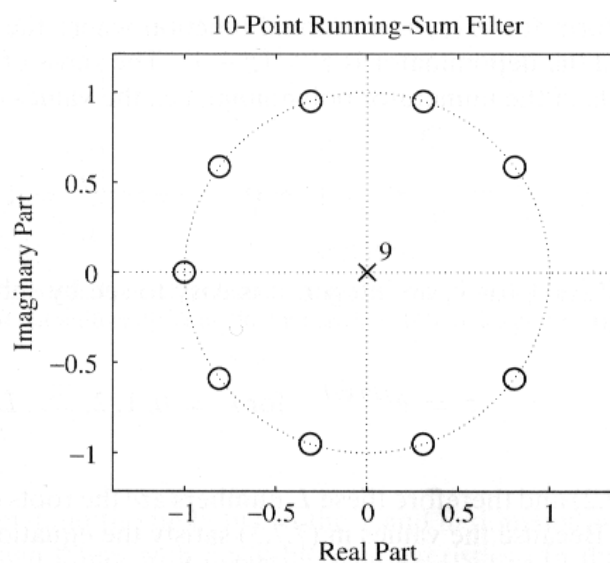


Figure 7.9. Pole-zero distribution of a running-sum filter.

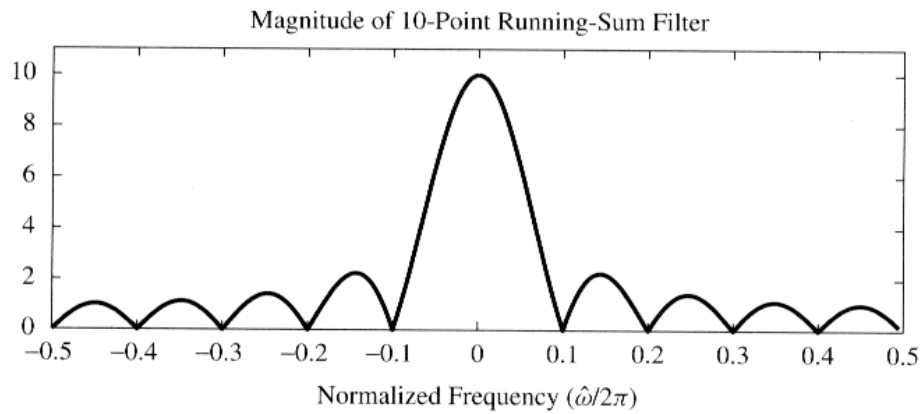


Figure 7.10. Magnitude response of a running-sum filter.

For a length-10 filter, the system function is

$$H(z) = \sum_{k=0}^9 z^{-k} = \frac{1 - z^{-10}}{1 - z^{-1}}$$

The pole-zero diagram is shown in Figure 7.9 and the corresponding frequency response in Figure 7.10. The numerator contains the ten roots of unity, but the zero at $z=1$ is cancelled. Thus nine zeros remain around the unit circle, while the missing zero at $z=1$ allows a large response at zero frequency.

Interpretation. For $L=10$, zeros occur at $\omega = 2\pi k/10$ for $k=1, \dots, 9$. Thus the filter completely rejects complex sinusoids at those frequencies, while the missing zero at $z=1$ leaves the DC gain equal to 10.

$$H(1) = \sum_{k=0}^9 1 = 10$$

$$H(e^{j2\pi k/10}) = 0, \quad k = 1, \dots, 9$$

7.7.2 A Complex Band-Pass Filter

Shifting the passband away from zero produces a filter that passes a narrow band of frequencies: a band-pass filter.

The system function is

$$H(z) = \prod_{\substack{k=0 \\ k \neq k_0}}^{L-1} (1 - e^{j2\pi k/L} z^{-1})$$

where k_0 identifies the omitted root at $z=e^{j2\pi k_0/L}$. For $L=10$ and $k_0=2$, the passband is shifted to $\omega = 2\pi k_0/L = 0.4\pi$.

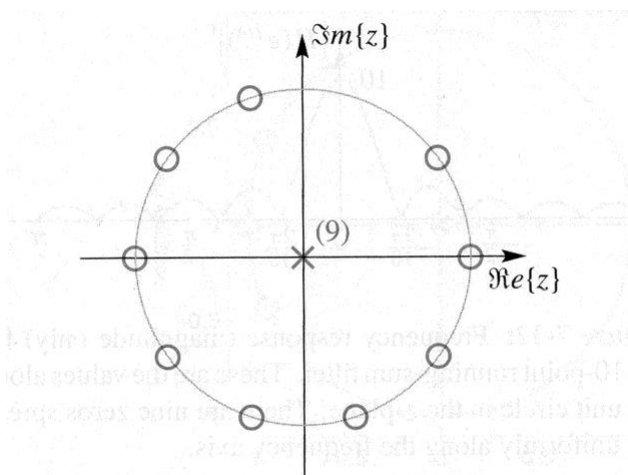


Figure 7.13. Pole-zero distribution of a length-10 complex band-pass filter.

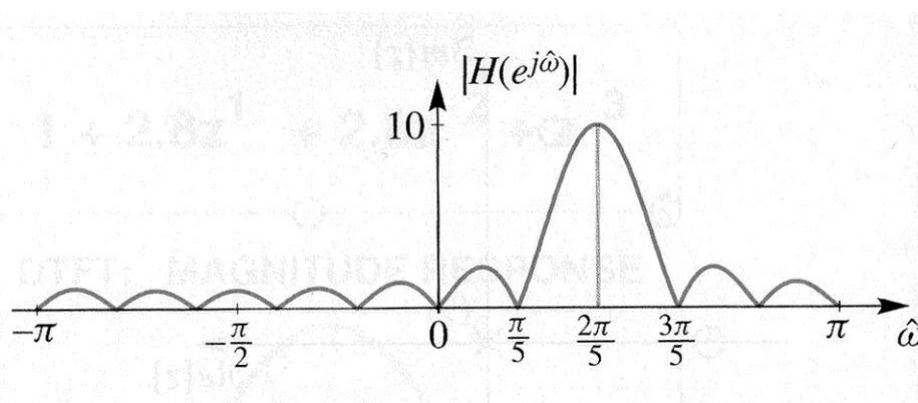


Figure 7.14. Magnitude response of the filter in Figure 7.13.

7.7.3 A Band-Pass Filter with Real Coefficients

The coefficients can be made real by taking the real part of the coefficients of the preceding complex filter. The k th coefficient is

$$b_k = \cos\left(\frac{2\pi k_0 k}{L}\right), \quad k = 0, 1, \dots, L-1$$

With these real coefficients, the new band-pass filter is the sum of two complex band-pass filters.

$$\begin{aligned} H(z) &= \sum_{k=0}^{L-1} \cos\left(\frac{2\pi k_0 k}{L}\right) z^{-k} \\ &= \frac{1}{2} \sum_{k=0}^{L-1} e^{j2\pi k_0 k/L} z^{-k} + \frac{1}{2} \sum_{k=0}^{L-1} e^{-j2\pi k_0 k/L} z^{-k} \\ &= H_1(z) + H_2(z) \end{aligned}$$

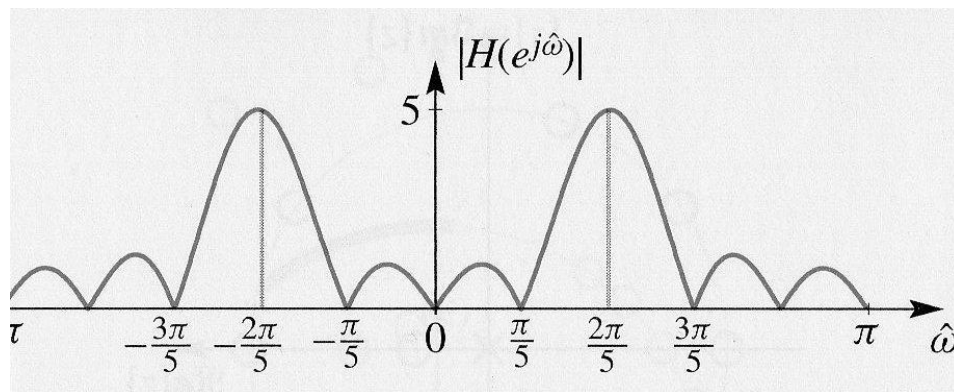


Figure 7.15. Magnitude response of a length-10 real band-pass filter.

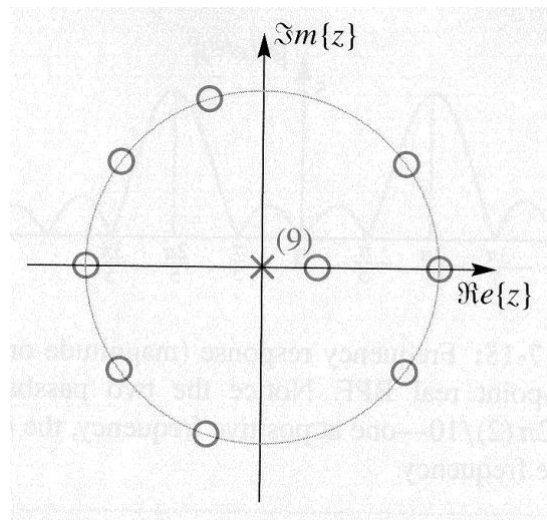


Figure 7.16. Pole-zero distribution of a length-10 real band-pass filter.

7.8 Practical Design of Band-Pass Filters

Characterising filters through their poles and zeros is useful in filter design and implementation. The z-transform converts problems involving convolution and frequency responses into algebraic operations based on polynomial multiplication and factorisation.

More complex filters can be designed with functions such as `remez` and `fir1`. These programs specify the desired passband and stopband ranges and calculate filter coefficients that approximate unit gain in the passband and zero gain elsewhere. Figures 7.17–7.19 show one example.

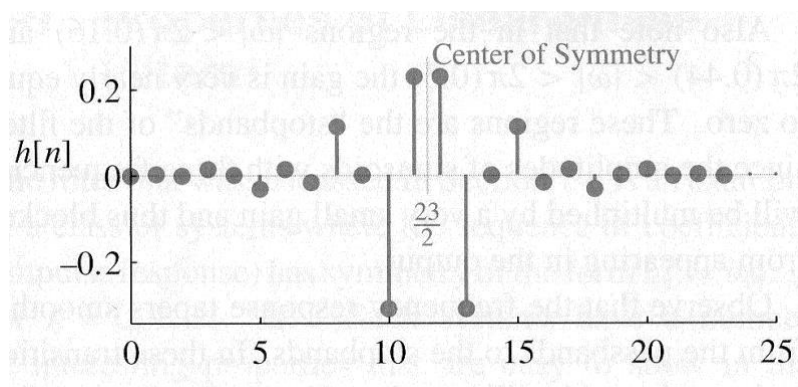


Figure 7.17. Impulse response of a length-24 band-pass filter designed with MATLAB `fir1`.

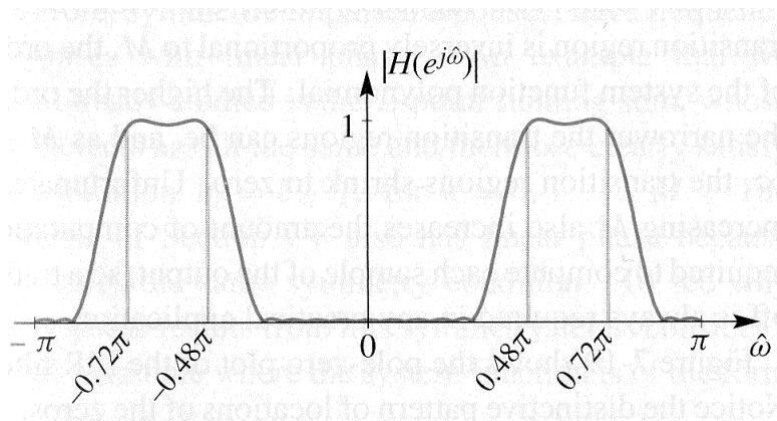


Figure 7.18. Magnitude response of the length-24 band-pass filter in Figure 7.17.

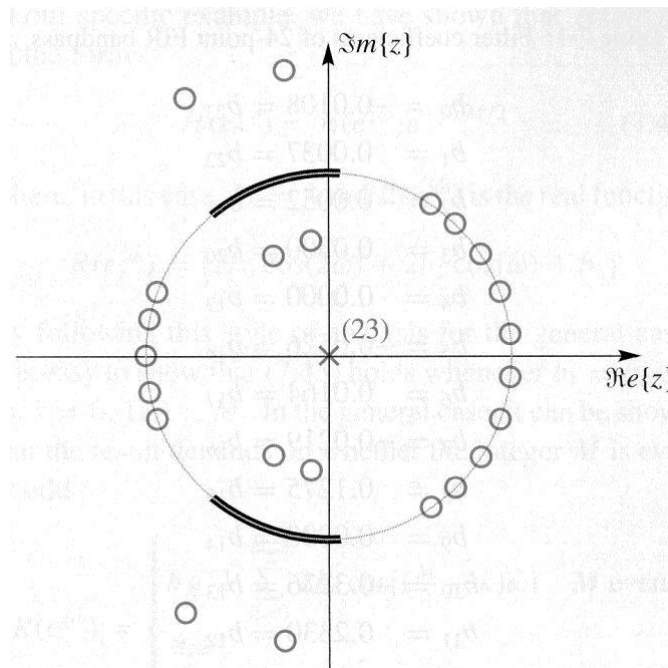


Figure 7.19. Pole-zero distribution of the length-24 band-pass filter in Figure 7.17.

7.9 Linear-Phase Properties of FIR Filters

FIR systems with symmetric coefficients, and therefore symmetric impulse responses, have a linear-phase frequency response.

Consider a fourth-order filter with system function

$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + b_1 z^{-3} + b_0 z^{-4}$$

Using coefficient symmetry, $H(z)$ can be written as

$$H(z) = [b_0(z^2 + z^{-2}) + b_1(z + z^{-1}) + b_2]z^{-2}$$

For a higher-order symmetric filter there would be more terms of the form $(z^k + z^{-k})$. Substituting $z = e^{j\omega}$ converts each symmetric pair into a cosine term, giving

$$H(e^{j\hat{\omega}}) = [2b_0 \cos(2\hat{\omega}) + 2b_1 \cos(\hat{\omega}) + b_2]e^{-j2\hat{\omega}}$$

7.10 Key points

- The z-transform represents a sequence or impulse response as a polynomial or rational function of z^{-1} .
- A delay of one sample corresponds to multiplication by z^{-1} .

- Convolution in the time domain corresponds to multiplication in the z domain.
- The frequency response is obtained by evaluating the system function on the unit circle, $z=e^{j\omega}$.
- Zeros on the unit circle suppress sinusoids at the corresponding angles; poles indicate locations where the system function becomes unbounded.
- Factorisation decomposes a higher-order system into cascaded lower-order systems, while inverse filtering corresponds to division of system functions.
- Symmetric FIR coefficients produce a linear-phase frequency response.

7.11 Notation

$x[n]$	Discrete-time signal or input sequence
$h[n]$	Impulse response of an LTI system
$y[n]$	Output sequence
$X(z)$	z-transform of $x[n]$
$H(z)$	System function or transfer function
$Y(z)$	z-transform of $y[n]$
z	Complex z-plane variable
z^{-1}	One-sample delay factor
ω	Normalized radian frequency in radians per sample
$H(e^{j\omega})$	Frequency response evaluated on the unit circle
z_0	A zero of $H(z)$
p_0	A pole of $H(z)$
$*$	Convolution operator
L	Filter length
M	Filter order